

Homework 8

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (5-1). Let f be defined for all real numbers. Assume that for every pair of real numbers x and y ,

$$|f(x) - f(y)| \leq (x - y)^2.$$

Prove that f is a constant function.

Proof. If there exists $x < y$ such that $f(x) \neq f(y)$, then take a positive integer N such that $(y - x)^2 < N |f(y) - f(x)|$. Divide the interval $[x, y]$ into N equal subintervals, and let $x_0 = x, x_1, \dots, x_N = y$ be the endpoints of these subintervals, such that $x_k - x_{k-1} = (y - x)/N := \Delta$. Then for any $1 \leq k \leq N$,

$$|f(x_k) - f(x_{k-1})| \leq (x_k - x_{k-1})^2 = \Delta^2.$$

Summing these inequalities from $k = 1$ to $k = N$, we get

$$|f(y) - f(x)| \leq \sum_{k=1}^N |f(x_k) - f(x_{k-1})| \leq N \Delta^2 = \frac{(y - x)^2}{N}.$$

This contradicts our assumption that $(y - x)^2 < N |f(y) - f(x)|$. Therefore, $f(x) = f(y)$ for all $x, y \in \mathbb{R}$, and f is a constant function. $\square$

Problem 2 (5-4). Let $C_0, \dots, C_n$ be real constants, and suppose

$$C_0 + \frac{C_1}{2} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0.$$

Prove that the equation

$$C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real root in $(0, 1)$.

Proof. Define

$$f(x) = C_0 x + \frac{C_1}{2} x^2 + \dots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}.$$

Then $f(0) = 0$ and $f(1) = C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$. By Rolle's theorem, there exists some $c \in (0, 1)$ such that $f'(c) = 0$. Note that

$$f'(x) = C_0 + C_1x + \cdots + C_{n-1}x^{n-1} + C_nx^n.$$

Therefore, $f'(c) = 0$ implies that the equation $C_0 + C_1x + \cdots + C_{n-1}x^{n-1} + C_nx^n = 0$ has at least one real root in $(0, 1)$. $\square$

Problem 3 (5-5). Let f be defined and differentiable for all $x > 0$, and assume that $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$. Define

$$g(x) = f(x+1) - f(x).$$

Prove that $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.

Proof. By the mean value theorem, for each $x > 0$, there exists some $c_x \in (x, x+1)$ such that

$$g(x) = f(x+1) - f(x) = f'(c_x)(x+1-x) = f'(c_x).$$

Since $c_x \rightarrow +\infty$ as $x \rightarrow +\infty$ and $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$, we have $g(x) = f'(c_x) \rightarrow 0$ as $x \rightarrow +\infty$. $\square$

Problem 4 (5-6). Assume:

- (a) f is continuous on $x \geq 0$;
- (b) $f'(x)$ exists for $x > 0$;
- (c) $f(0) = 0$;
- (d) f' is increasing.

Let

$$g(x) = \frac{f(x)}{x} \quad (x > 0).$$

Prove that g is increasing.

Proof. Suppose the contrary, that is, there exist $0 < a < b$ such that $g(a) > g(b)$. By the mean value theorem, there exists some $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b-a).$$

Since $g(a) > g(b)$, we have

$$f'(c) = \frac{f(b) - f(a)}{b-a} = \frac{g(b)b - g(a)a}{b-a} < \frac{g(a)b - g(a)a}{b-a} = g(a).$$

By the mean value theorem again, there exists some $d \in (0, a)$ such that

$$f(a) - f(0) = f'(d)(a-0) \implies f'(d) = g(a).$$

Since f' is increasing and $d < c$, we have $f'(d) \leq f'(c)$, which contradicts $f'(d) = g(a) > f'(c)$. Therefore, g is increasing. $\square$

Problem 5 (5-9). Let f be a continuous real-valued function on $\mathbb{R}$. Assume that $f'(x)$ exists for every $x \neq 0$, and that $f'(x) \rightarrow 3$ as $x \rightarrow 0$. Question: does $f'(0)$ necessarily exist?

Solution. Yes. We will prove that

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 3.$$

First of all, by $\lim_{x \rightarrow 0} f'(x) = 3$, we know that f is differentiable at some neighbourhood of 0. For any $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $0 < |x| < \delta$, we have $|f'(x) - 3| < \varepsilon$. Then for all $0 < |x| < \delta$, by the mean value theorem, there exists some $c_x \in (0, x)$ or $(x, 0)$ such that

$$\frac{f(x) - f(0)}{x} = f'(c_x) \in (3 - \varepsilon, 3 + \varepsilon).$$

This implies that $\lim_{x \rightarrow 0} (f(x) - f(0))/x = 3$, and thus $f'(0)$ exists and equals 3.

Problem 6 (5-11). Let f be defined in a neighborhood of x , and assume $f''(x)$ exists. Prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x).$$

Give an example showing that this limit may still exist even when $f''(x)$ does not exist.

Problem 7 (5-17). Let f be a real-valued function on $[-1, 1]$ that is three times differentiable, and suppose

$$f(-1) = 0, \quad f(0) = 0, \quad f(1) = 1, \quad f'(0) = 0.$$

Prove that there exists some $x \in (-1, 1)$ such that

$$f^{(3)}(x) \geq 3.$$

Note: for the function $\frac{1}{2}(x^3 + x^2)$, equality is attained.

Problem 8 (5-22). Let f be a real function on $(-\infty, \infty)$. If $f(x) = x$, then x is called a fixed point of f .

(a) If f is differentiable and $f'(t) \neq 1$ for every real t , prove that f has at most one fixed point.

(b) Prove that

$$f(t) = t + (1 + e^t)^{-1}$$

has no fixed point, although for every real t one has $0 < f'(t) < 1$.

(c) However, if there is a constant $A < 1$ such that for every real t ,

$$|f'(t)| \leq A,$$

prove that f has a fixed point x , and moreover

$$x = \lim_{n \rightarrow \infty} x_n,$$

where x_1 is any real number and for $n = 1, 2, 3, \dots$,

$$x_{n+1} = f(x_n).$$

(d) Prove that the method in (c) can be implemented along the zig-zag path

$$(x_1, x_2) \rightarrow (x_2, x_2) \rightarrow (x_2, x_3) \rightarrow (x_3, x_3) \rightarrow (x_3, x_4) \rightarrow \cdots .$$